MILITARY SPECIFICATION

CLOTH, FLANNEL, WOOL

This specification is approved for use by Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements for one type of wool flannel cloth.

2. APPLICABLE DOCUMENTS

*2.1 Government documents

*2.1.1 Specifications, standards, and handbooks Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATIONS

FEDERAL

PPP-P-1132 - Packaging and Packing of Woolen, Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics

MILITARY

MIL-C-43665 - Cloth, Wool; Mothproofing Treatment of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-655 - Provisions for Evaluating Quality of Cloth, Wool and Wool Blends

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.1.2 Other government documents, drawings, and publications The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

RULES AND REGULATIONS UNDER THE WOOL PRODUCT LABELING ACT OF 1939

(Application for copy should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications The following document(s) form a part of this specification to the extent specified herein. The issues of documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

US DEPARTMENT OF AGRICULTURE
Methods of Test for Grade of Wool

(Application for copies should be addressed to the US Department of Agriculture, Agricultural Marketing Service, Washington, DC 20402).

3. REQUIREMENTS

*3.1 Standard sample The finished cloth shall match the standard sample for shade, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article When specified, the contractor shall furnish a sample of cloth for first article inspection and approval (see 4.3 and 6.2).

3.3 Materials

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3.3.1 Wool stock The wool stock shall be fleece, or pulled wool or both, not lower in grade than 60's US Standard (see 4.4.1.1). The use of reprocessed, reused or waste wool is prohibited. The wool shall be free from vegetable matter, or other foreign matter.

3.3.2 Yarn The wool stock shall be thoroughly scoured, blended, carded and evenly spun on the woolen system into singles yarn for both warp and filling to meet the requirements described herein.

NOTE: Selvage yarns - To prevent edges of cloth from building up when rolled, selvage yarns may be of a finer count than those used in the body of the cloth.

3.4 Weave The weave shall be a 2 up, 2 down right hand twill.

*3.5 Physical requirements The finished cloth shall conform to the requirements listed in Table I when tested as specified in 4.5.

Table I - Physical requirements

Weight, oz (g) per linear yd of 56 inches		Yarns/inch (per cm)		Breaking Strength lbs. (Newtons)		Shrinkage sponging (percent)	
Min	Max	Minimum		Minimum		Maximum	
		Warp	Filling	Warp	Filling	Warp	Filling
11 (311.7)	12 (340.1)	56 (22.0)	48 (18.9)	35 (156)	35 (156)	4.5	4

3.5.1 Width Unless otherwise specified (see 6.2), the width shall be not less than 56 inches (142.2cm) exclusive of selvage.

3.6 Wool content The finished cloth shall contain not less than 95.0 percent wool when tested as specified in 4.5.

3.7 Color The color shall be Blue 3346 and shall be produced by stock dyeing with suitable dyestuffs of required fastness properties to match the standard sample.

*3.7.1 Matching The color of the finished cloth shall match the standard sample under artificial daylight, having a color temperature of 7500 $\pm$ 500 Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2850 $\pm$ 100 Kelvin.

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3.7.2 Colorfastness The finished cloth shall show colorfastness to light, wet-dry cleaning, perspiration, laundering and crocking equal to or better than the standard sample when tested as specified in 4.5. When no standard sample is established or designated as applicable to colorfastness, the finished cloth shall show a minimum of "good" fastness to light, wet-dry cleaning, perspiration and laundering and shall show a Munsell Value for crocking not lower than 8.5 when tested as specified in 4.5.

3.8 Finish The finished cloth shall have a flannel finish, be well pressed and have a lustrous face, nap and hand equal to the standard sample.

3.9 Length and put-up Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths of not less than 40 yards (36.5m) nor more than 100 yards (91.4m) in length. Each length shall be put-up in full width rolls as specified in PPP-P-1132.

3.10 Wool content label Cloth manufactured under this specification shall be labeled in accordance with the Wool Products Labeling Act of 1939 (see 2.1.2).

3.11 pH The pH value of the water extract of the finished cloth shall be not less than 4.0 nor more than 8.0 when tested as specified in 4.5.

3.12 Mothproofing The cloth shall be given a mothproofing treatment in accordance with MIL-C-43665 (see 4.5).

3.13 Marking The back of the cloth shall be marked "Back" at both ends of each roll with letters not less than 1/2 inch (12.7mm) in height (see 4.4.2). The marking shall be clearly legible in any indelible marking used commercially and shall not show through on the face of the cloth.

*3.14 Workmanship The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards ($83.6m^2$), when calculated as specified in Section 4, shall not exceed the established maximum point value. Each defect shall be marked with a 1 to 1 1/2 inch (2.5 to 3.8cm) long string. The string shall be sewn into the selvage near the defect. A red string shall represent a three or four point defect and a white string shall represent a one or two point defect (see 4.4.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

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4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspections The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article inspection When required, the first article of finished cloth shall be visually examined for color, finish and appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.5.

* 4.4 Quality conformance inspection Inspection shall be performed in accordance with MIL-STD-655, except where otherwise indicated.

4.4.1 Component and material inspection In accordance with 4.1 above, components and materials shall be visually examined and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1.1 Wool The wool shall be visually examined for grade in scoured form, prior to dyeing, picking or carding by comparison with the applicable US Standard (see 2.2). The examination for grade shall be performed on a composite sample of 1 pound (0.45Kg) of scoured wool for each 10,000 pounds (4500Kg) or fraction thereof in the lot. The composite sample shall be selected from not less than 10 percent of the bags/bales in the lot. In the event of dispute resulting from above comparisons, the grade shall be determined by a width method (wedge). The lot shall be unacceptable if one or more sample units fail to meet the requirements of 3.3.1. Each lot shall consist of one grade of wool only.

* 4.4.1.2 In-process inspection Inspection shall be made at any point or during any phase of the manufacturing process to assure that only approved wools are used (see 3.3.1). The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance.

* 4.4.2 Examination of the end item Examination of the end item for yard-by-yard, length, back marking, compliance with the Wool Product Labeling Act, and for defect marking (Stringing of defects) shall be as specified in MIL-STD-655. The fabric quality for the finished cloth shall be quality level 10.

4.4.2.1 Limits of slubs and knots Only slubs and knots which exceed "C" for slubs and "D" for knots on the Sears Fabric Defect Scale (see 6.4) shall be scored.

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4.4.2.2 Examination for shade and finish, individual rolls Each roll in the lot shall be examined visually for shade match (see 3.7), and finish (see 3.8). A roll shall be unacceptable if it fails to meet the requirements for shade match or finish as applicable. The sample unit shall be a 4-inch by 20-inch (10.1 by 50.8cm) swatch of the cloth. A sample unit shall be drawn from each roll in the lot.

4.4.3 Examination of packaging requirements An examination shall be made in accordance with the provisions of PPP-P-1132 to determine that packaging, packing and marking comply with the Section 5 requirements of this specification.

4.5 Testing of the end item The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table II, shall be followed. The physical and chemical values specified in Section 3, except where otherwise indicated, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 3 1/2 yds (3.2m), full width of the finished cloth. The lot size shall be expressed in units of one yard (0.9m). The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II. Test methods

Characteristics	Requirement paragraph	Test method
Wool content	3.6	2101
Weave	3.4	Visual 1/
Weight	3.5	5041
Yarns per inch:		
Warp	3.5	5050
Filling	3.5	5050
Breaking strength:		
Warp	3.5	5100
Filling	3.5	5100
Shrinkage sponging	3.5	5590
Colorfastness to:		
Light	3.7.2	5660
Wet-dry cleaning	3.7.2	5622
Perspiration	3.7.2	5680
Laundering	3.7.2	5614
Crocking	3.7.2	5651
pH	3.11	2811
Mothproofing	3.12	2/

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1/ One determination shall be made on each sample unit and the results reported as "pass" or "fail".
2/ As specified in MIL-C-43665.

5. PACKAGING

5.1 Preservation-packaging Packaging shall be level A or C (see 6.2).

5.1.1 Levels A and C The cloth, put-up as specified (see 3.9) shall be packaged in accordance with the applicable requirements of PPP-P-1132.

5.2 Packing Packing shall be level A, B, or C (see 6.2).

5.2.1 Levels A, B and C The cloth shall be packed in accordance with the applicable requirements of PPP-P-1132.

5.3 Marking In addition to any special marking required in the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1132.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in the manufacture of uniform items worn by Navy personnel.

* 6.2 Ordering data Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Width of cloth required when other than specified (see 3.5.1).
c. Length required when other than specified (see 3.9).
d. Levels of packaging and packing required (see 5.1 and 5.2).
e. When a first article inspection is required, the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, quantity, and testing and approval of the first article.

6.3 Standard sample For access to the standard samples, address the procuring activity issuing the invitation for bids (see 3.1).

* 6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Co, Department 817, (ATTN: BSC-23-29), Sears Tower, Chicago, IL 60684.

6.5 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-16291G(NU)	Cloth, Flannel, Wool

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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